UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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PART A (80 MARKS)

QUESTION 1

a) State two reasons why software insecurity arises, according to the lecture.

(2 marks)

b) Explain three Windows security mechanisms or components described in the lecture and, for each, one security concern that arises from it.

(6 marks)

QUESTION 2

a) Explain how a substitution cipher works and what serves as its key.

(2 marks)

b) Decrypt the following ciphertext using the Vigenere cipher with the key CIPHER. Use the standard 26 x 26 Vigenere tableau, writing the key over the message and repeating it as necessary.

UMRBVVNWVPREQE

(6 marks)

Vigenere tableau (A = 0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

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QUESTION 3

a) List two characteristics of Trojan horses as a type of malware.

(2 marks)

b) State six techniques by which malware attacks systems or propagates, including how a virus infects executable files and how a worm spreads.

(6 marks)

QUESTION 4

a) Explain the Chinese Wall model's purpose and its two access rules.

(2 marks)

b) [COMPUTED PLACEHOLDER - acm, 6 marks]

(6 marks)

QUESTION 5

a) State the two stages of the biometric authentication process.

(2 marks)

b) Discuss six aspects of biometric mechanisms and technologies, covering their definition, categories, and the strengths and weaknesses of specific biometric methods.

(6 marks)

QUESTION 6

a) State TWO (2) differences between morals and laws.

(2 marks)

b) Explain how a professional code of ethics guides computing professionals, referring to the ACM code and ACM professional conduct.

(6 marks)

QUESTION 7

a) Explain how NIDS sensors should be positioned in a network and give two typical sensor locations.

(2 marks)

b) Refer to your answer in (a). Explain six preventive network security mechanisms, other than the NIDS, that an organization can deploy to protect its network, and state what each mechanism does.

(6 marks)

QUESTION 8

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a) List TWO (2) password security controls used to protect user accounts.

(2 marks)

b) Describe six (6) aspects of Kerberos and single sign-on mechanisms, covering their motivation, design criteria, login procedure and alternatives.

(6 marks)

QUESTION 9

a) Explain how the classes of threats affect confidentiality, integrity and availability.

(2 marks)

b) State six facts about the Samy worm and the XSS attack it used, covering how the script was injected, how it spread, and the outcome.

(6 marks)

QUESTION 10

a) Explain two limitations of passive intrusion detection systems in defending against attacks.

(2 marks)

b) Explain six points about threat classes and the countermeasures used against attacks, covering how threats differ from vulnerabilities and attacks, how misuse detection counters known attacks, and how anomaly-based prediction counters unknown activity.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A consultant at a security firm must advise a client on how to represent and enforce access rights over shared documents and executables. The client wants a structure that is easy to verify, yet also wants to know the drawbacks of each representation before choosing one.

a) Explain the access control matrix, access control list and capability structures as ways of representing access rights, including one advantage and one disadvantage of each.

(5 marks)

b) State the four goals of access control that the consultant should consider when advising the client, and explain the role of a reference monitor in realizing the client's chosen access control policy.

(5 marks)

QUESTION 2

A hospital wants to prove to auditors that its servers boot only approved software. The vendor proposes a tamper-resistant chip on each motherboard that records what loads at

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startup, so that the final platform state can be checked against a known-good value.

a) Referring to the hospital proposal, explain how a TPM and a chain of trust together let the platform's boot state be verified as trusted.

(5 marks)

b) Discuss the NGSCB/Nexus components that the vendor in the scenario could use to isolate the hospital's trusted software from the ordinary operating system, and explain how each component contributes to that isolation.

(5 marks)

END OF QUESTION PAPER

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